THE CHOW GROUP IN DIMENSION K IS THE IMAGE OF THE K-GROUP OF SHEAVES SUPPORTED IN DIMENSION AT MOST K INSIDE THE K-GROUP OF SHEAVES SUPPORTED IN DIMENSIONS K+1 AND K

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 02SC (sources/stacks-chow/pinned-original-source.tex line 18509).

Allowed standard prerequisites (editorial).

- Identification of Z_k(X) with the K-group of the quotient of the category of sheaves supported in dimension at most k. Same chapter, Lemma labelled lemma-cycles-k-group (source line 3757) and the construction of Lemma labelled lemma-from-chow-to-K (source line 3934), invoked at statement lines 18517 and proof lines 18527-18529. Fix the map whose bijectivity modulo the subgroup B_k(X) is the content of the lemma, and identify the source with the cycle group.
- K-groups of a Serre subcategory and its quotient: killing a class and realising it by a (2, 1)-periodic complex. Homology, Lemma labelled homology-lemma-serre-subcategory-K-groups (Stacks tag 02MX), parts (1) and (2), invoked at proof lines 18545-18549. Supplies both the passage from the vanishing of [A] - [B] in the larger quotient to the existence of A', B' supported in dimension at most k-1 and the realisation of the resulting class by a (2, 1)-periodic complex; the cycle comparison is done in the counted span.
- Rational equivalence of the k-cycles of the two cohomology sheaves of such a complex. Same chapter, Lemma labelled lemma-maps-between-coherent-sheaves (source line 18417), invoked at proof line 18554. Supplies [A + A']_k ~ rat [B + B']_k for a (2, 1)-periodic complex satisfying the support conditions; the reduction of the K-theoretic statement to this input is inside the counted span.
- The base situation (S, delta) and the dimension conventions for supports. Same chapter, Situation labelled situation-setup (source line 1372), invoked

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in the statement. Fixes the delta-dimension, the categories Coh_{<= k}(X) and the cycle group Z_k(X) in which the statement is formulated.

Source reference. Original definition 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Same-chapter pointer into the author environment situation at source line 1372; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FDS: sources/stacks-chow/pinned-original-source.tex, source lines 3934–3934. Same-chapter pointer into the author environment lemma at source line 3934; invoked at proof line 18529 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02S9: sources/stacks-chow/pinned-original-source.tex, source lines 3757–3757. Same-chapter pointer into the author environment lemma at source line 3757; invoked at proof line 18527 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02MX: sources/stacks-chow/tags.txt, source lines 3460–3460. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 18546 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02SB: sources/stacks-chow/pinned-original-source.tex, source lines 18417–18417. Same-chapter pointer into the author environment lemma at source line 18417; invoked at proof line 18554 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be a scheme locally of finite type over S. The map*

$$\mathrm{CH}_k(X) \longrightarrow K_0(Coh_{\leq k+1}(X)/Coh_{\leq k-1}(X))$$

from Lemma 0FDS induces a bijection from $\mathrm{CH}_k(X)$ onto the image $B_k(X)$ of the map

$$K_0(Coh_{\leq k}(X)/Coh_{\leq k-1}(X)) \longrightarrow K_0(Coh_{\leq k+1}(X)/Coh_{\leq k-1}(X)).$$

Proof. By Lemma 02S9 we have $Z_k(X) = K_0(Coh_{\leq k}(X)/Coh_{\leq k-1}(X))$ compatible with the map of Lemma 0FDS. Thus, suppose we have an element $[A] - [B]$ of $K_0(Coh_{\leq k}(X)/Coh_{\leq k-1}(X))$ which maps to zero in $B_k(X)$, i.e., maps to zero in $K_0(Coh_{\leq k+1}(X)/Coh_{\leq k-1}(X))$. We have to show that $[A] - [B]$ corresponds to a cycle rationally equivalent to zero on X. Suppose $[A] = [\mathcal{A}]$ and $[B] = [\mathcal{B}]$ for some coherent sheaves $\mathcal{A}, \mathcal{B}$ on X supported in δ-dimension $\leq k$. The assumption that $[A] - [B]$ maps to zero in the group $K_0(Coh_{\leq k+1}(X)/Coh_{\leq k-1}(X))$ means that there exists coherent sheaves $\mathcal{A}', \mathcal{B}'$ on X supported in δ-dimension $\leq k-1$ such that $[\mathcal{A} \oplus \mathcal{A}'] - [\mathcal{B} \oplus \mathcal{B}']$ is zero in $K_0(Coh_{k+1}(X))$ (use part (1) of Homology, Lemma 02MX). By part (2) of Homology, Lemma 02MX this means there exists a $(2,1)$-periodic complex $(\mathcal{F}, \varphi, \psi)$ in the category $Coh_{\leq k+1}(X)$ such that $\mathcal{A} \oplus \mathcal{A}' = H^0(\mathcal{F}, \varphi, \psi)$ and $\mathcal{B} \oplus \mathcal{B}' = H^1(\mathcal{F}, \varphi, \psi)$. By Lemma 02SB this implies that

$$[\mathcal{A} \oplus \mathcal{A}']_k \sim_{rat} [\mathcal{B} \oplus \mathcal{B}']_k$$

This proves that $[A] - [B]$ maps to a cycle rationally equivalent to zero by the map

$$K_0(Coh_{\leq k}(X)/Coh_{\leq k-1}(X)) \longrightarrow Z_k(X)$$

of Lemma 02S9. This is what we had to prove and the proof is complete. □